

Artificial Intelligence Project : 8-Queens Problem

A* vs Neural Networks

Abstract :

The popular 8-Queens problem is a great standard due to its nature - discrete search space and the possibility of board configurations which grow exponentially. This project develops and compares two distinct artificial intelligence algorithms. The classic A* with a custom admissible heuristic and a hybrid optimization approach using MLP Regressor to perform local search.

AI Concepts : A* Search, Admissible Heuristic, MLP Regressor

Team Contributions :

- Jay Mhaiskar: Advanced research of A* & Neural Networks, implemented the systematic A* Search algorithm, configured state expansion mechanics and defined admissible heuristics for partial placements. Assisted in blockers on neural network implementation for 8-Queens.
- Navya Sri Ambati : Adapted the neural network data pipeline, structured the NN guided local search stochastic escape mechanisms and handled report documentation and editing.

Introduction :

Historically classic constraint satisfaction problems have relied on systematic, exhaustive search algorithms to guarantee optimal solutions. However, as problem dimensions scale or state spaces become complex, traditional exact methods can encounter bottlenecks in large discrete domains. While standard uninformed search methods like breadth first search or depth first search explores states blindly informed search techniques such as A* utilize domain specific heuristics to systematically navigate toward goal states with high efficiency.

Modern artificial intelligence bridges symbolic search with sub-symbolic machine learning by using neural networks as evaluation functions. Instead of computing exact cost functions or relying only on manual heuristics a machine learning model can be trained to approximate the conflict density of a given state.

Related Works :

AI heavily used uninformed search, backtracking and constraint propagation to solve classical puzzles like 8-Queens. With heuristic search, algorithms like A* became AI curriculum staples, proving that distance-to-go estimates drastically

reduce expanded nodes. Meanwhile, local search algorithms showed that heuristic optimization could skip massive tree searches by operating directly on complete configurations, though they risk getting stuck in local minima. This project bridges classical search theory with modern ML applications.

Data Pipeline :

Phase 1 - We generated a robust synthetic training dataset of 20,000 board configurations by randomly sampling full 8-Queens states across the discrete space, where each board is represented as a list of length 8 with column indices mapping to row values (0 to 7)

Phase 2 - For every randomly generated board, we calculated the exact number of attacking queen pairs using a custom conflict evaluation function, mapping each board state directly to its integer conflict count to frame the task as a supervised regression problem.

Phase 3 - The input feature vector consists of the integer values representing the row positions of queens across columns 0 through 7. We split the dataset using an

80/20 train-test allocation, resulting in 16,000 training samples and 4,000 independent testing samples.

Artificial Intelligence Models :

- **A* Search Baseline Model** - Implemented as a systematic search on partial placement of queens as tuples. The cost function uses depth and heuristic uses the admissible estimate.
- **Neural Network (MLP Regressor)** - Built using MLP Regressor with 2 hidden layers (64, 32) , ReLU and Adam optimizer.

Error Analysis:

- **A* Search** - Reached a 100 % success with swift execution and minimal node expansion, incremental search is exceptionally efficient.
- **Neural Network** - Achieved a 5 % success across 20 runs. While the MLP Regressor learned general trends predicting precise integer conflict counts in a highly non-linear, discrete space introduces prediction noise.

Performance Analysis - Comparing A* against the Neural Networks local searches across 20 independent trials

Model	Success Rate	Average Runtime	Average States explored
A* Search	100 %	0.0073s	1,966 nodes
Neural Network	5 %	1.6302s	548,920 states

Conclusion :

Our results confirm that while neural networks can serve as flexible evaluators for heuristic search, unrefined regression models might struggle with discrete, highly sensitive landscapes like 8-Queens compared to exact systematic algorithms - A*.

Code : It can be found here @ [8-Queens with A* and Neural Networks](#)

References :

Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig